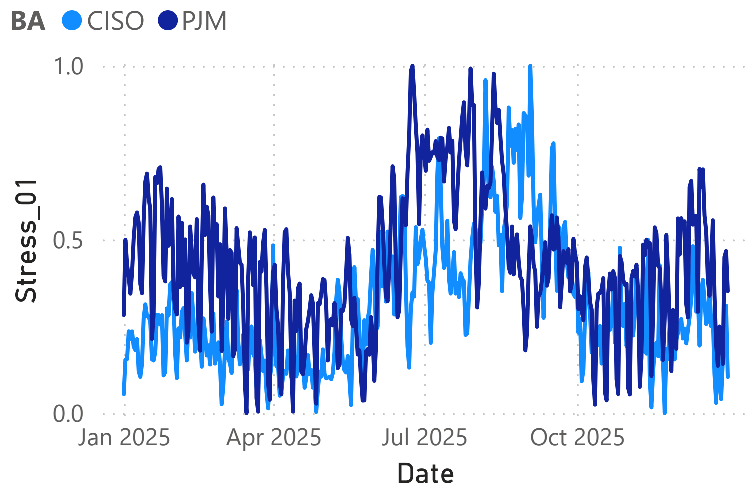
Electric Grid Stress Analysis - Overview

This dashboard visualizes a machine learning framework built on public grid data classifying daily stress regimes across multiple Balancing Authorities (BAs) using z-scores

Analytic Structure

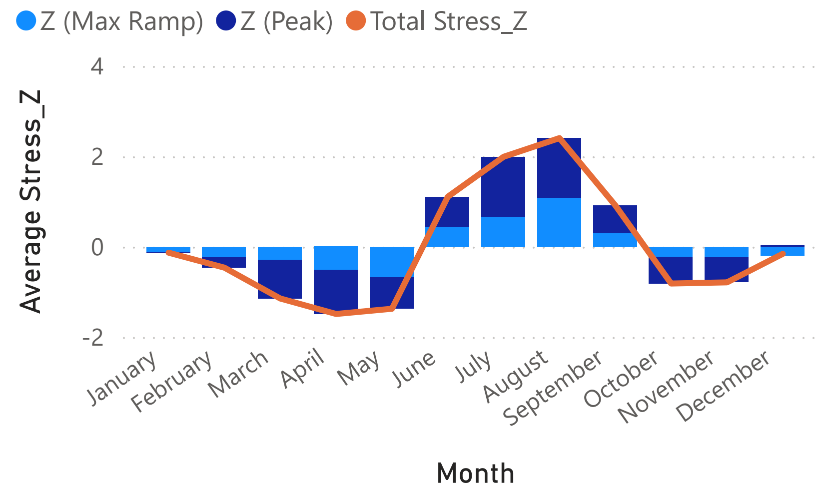
Stress_Z = Z_Peak + Z_Max_Ramp (mean = 0, value ≈ std dev)
Stress_01 = normalized Stress_Z (min = 0, max = 1)
Peak = Daily max power demand
Max_Ramp: Daily max power ramp between consecutive hours

Daily Stress_01



Both BAs follow similar but not identical stress trends throughout the year

Average Stress by Month



PJM stress is positive in winter months
PJM summer stress hits a few months earlier than CISO
Note: Stress by Month only informative by separated BAs

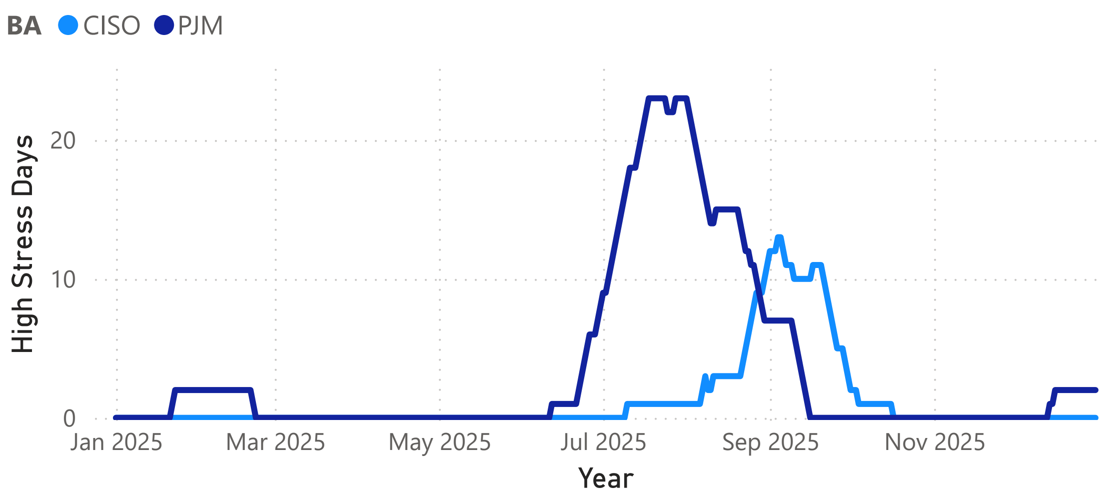
Balancing Authority (BA)

Select all | CISO | PJM

Most Stressful Days Table

BA	Stress_01	Date
CISO	0.82	August 23
CISO	0.82	August 25
CISO	0.83	August 27

Number of High Stress Days in the Previous 30 Days



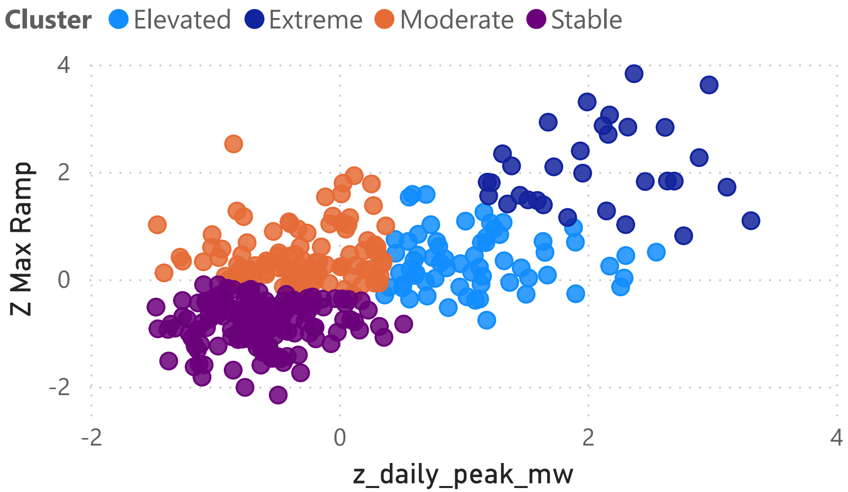
PJM has far more stressful days in the summer, and some in the winter

	Stress_Z (Max)	Stress_01 (Avg)	Most Stressed Day	High Stress Days
CISO	6.60	0.31	Sep 03	18
PJM	4.83	0.43	Jun 24	46

The stress trends seen in other figures are also demonstrated by these KPIs

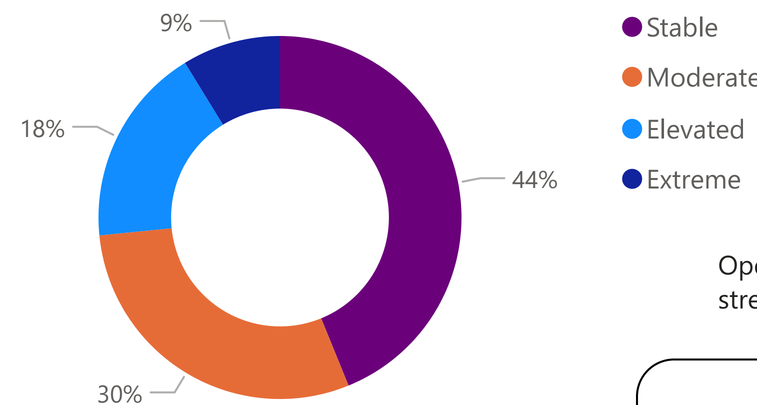
Electric Grid Stress Analysis - ML Clustering

Z-Score Clustering



Both BAs cluster similarly with instability increasing along the diagonal. CISO reaches higher extremes with more spread out datapoints

Frequency by Cluster



PJM contains more extreme days and CISO contains far more stable days - as to be expected with California's mild weather

Balancing Authority

CISO | PJM

Operators can use this calculator to determine how stressed their grid would be given specified conditions

Daily Stress Calculator

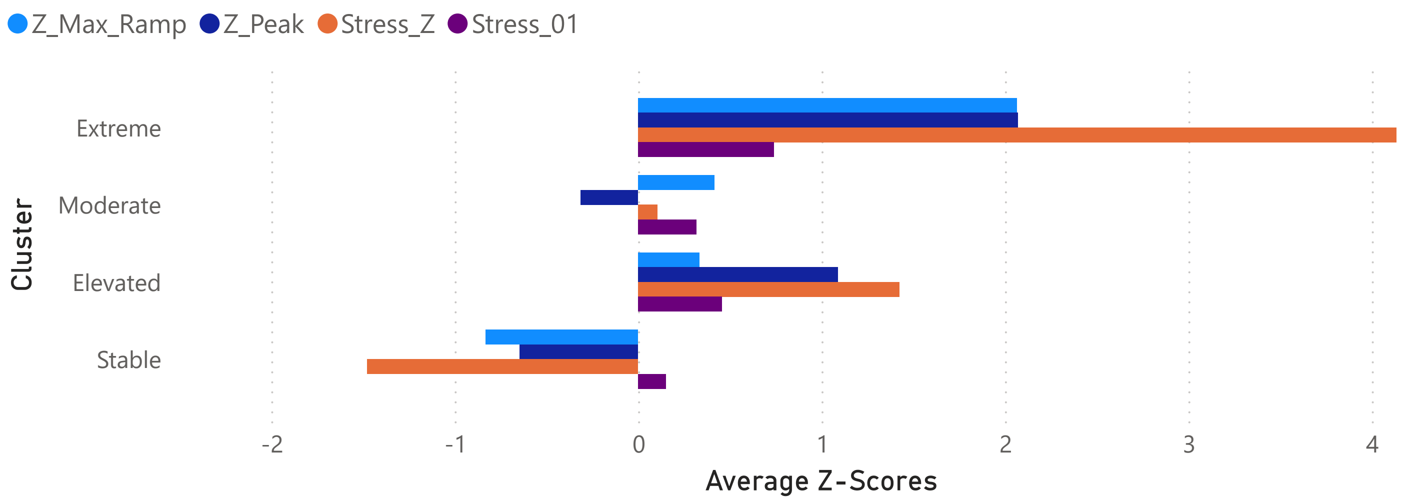
Peak Input (MW)	Max Ramp Input (MW)
35000	4000

Extreme Identified Cluster	**95%** Stress Percentile
3.81 Input Stress Z	**4.13** Cluster Mean Stress Z

Approximate 2025 Ranges (MW)

CISO Peak: 23,000 - 45,000 || CISO Ramp: 1,000 - 5,000
PJM Peak: 78,000 - 160,000 || PJM Ramp: 2,500 - 9,100

Average Z-Scores by Cluster

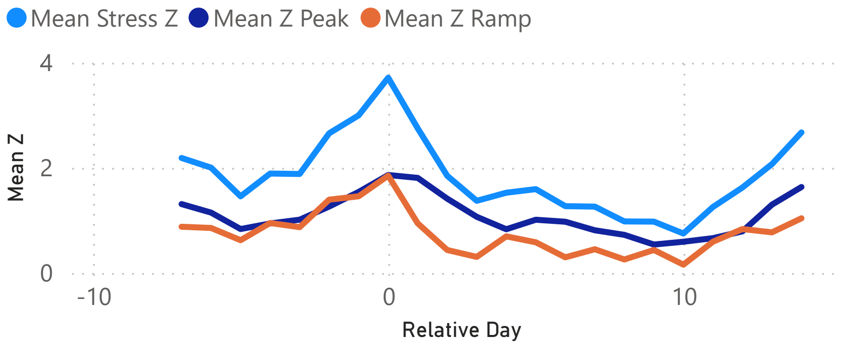


CISO reaches higher extreme stress scores due to higher ramp scores

Electric Grid Stress Analysis - Event Drilldown

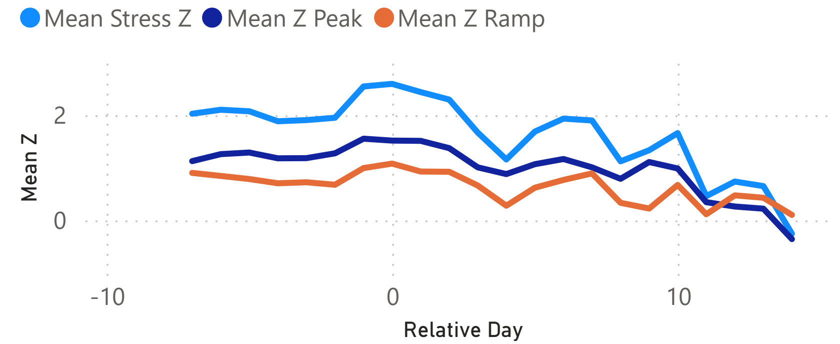
Composite events - An average of the extreme stress events. How stress builds and resolves around extreme events (day 0 = stress peak)

Composite Extreme Event (CAISO Summer)



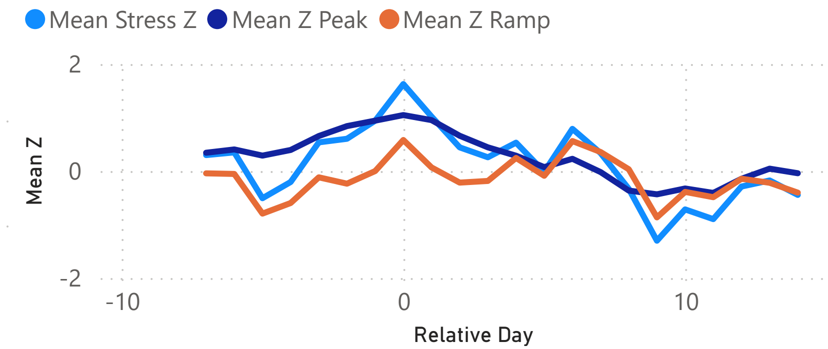
Ramp-dominated — solar drop-off drives the evening stress spike
• sharp ramp drives stress on extreme day when solar drop-off occurs

Composite Extreme Event (PJM Summer)



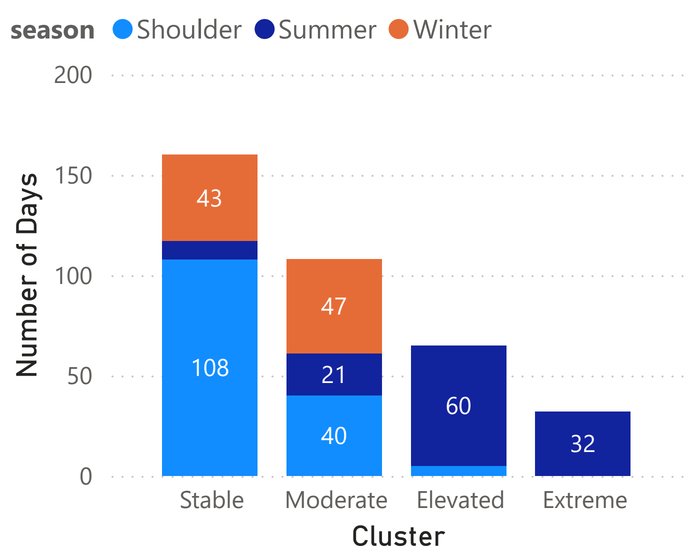
Load-dominated - slower build-up and drop off due to heat dome effect

Composite Extreme Event (PJM Winter)



Load-dominated — quick cold snap raises demand
• ramp signal is weak or negative

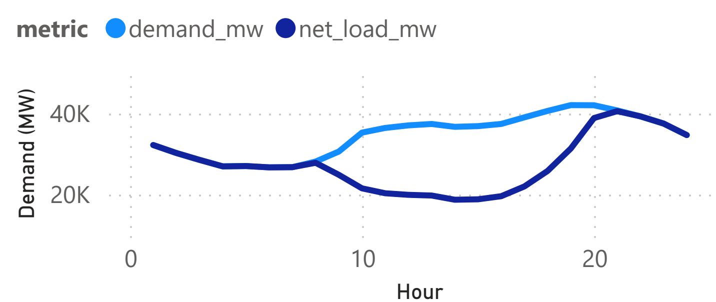
Number of Days by Cluster and Season



Above - PJM has more extreme days (including winter days)

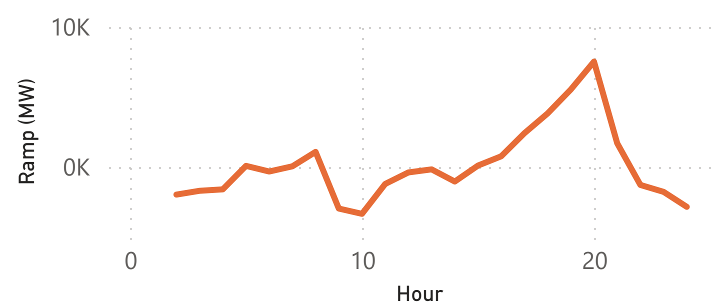
Right - Hourly differences by BA and season demonstrated
• i.e. CISO summer duck curve and PJM winter heating ramps

Hourly Demand on Extreme Day

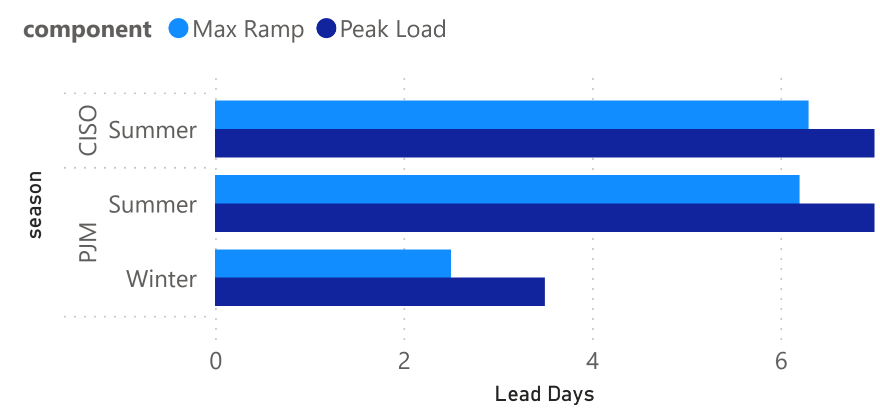


CISO | PJM | Summer

Hourly Ramp on Extreme Day



Lead Days to Extreme Event



Summer events give much more warning than winter PJM events

Extra Notes:
- All data is based on 2025, but analysis can be applied to other years
- CISO solar penetration is ~25% but PJM solar penetration is only ~5%